

## Tutorial Problems - APSC 171 - 2026

### Week 1 Material - Derivatives and Vectors

Some problems and solutions selected or adapted from Stewart Calculus.

1. <sup>1</sup> Find the derivatives of the following functions.

(a)  $f(t) = \sqrt{t^2 - 3t}$

(b)  $g(t) = \frac{1}{\sqrt{t^2 - 3t}}$

(c)  $h(x) = \frac{1}{1 + e^{-x}}$ .

(d)  $h(t) = \ln(8 - t^3)$ .

(e)  $f(x) = \ln(\cos^2(x) + 1) \cdot (e^x + \sin(x))^3$ .

2. <sup>2</sup> A particle is moving on a laboratory track with its position defined by  $r(t)$  as given below. **Compute the velocity function** for the particle.

(a)  $r(t) = \frac{5t^4 - 2t}{\sqrt{t}}$ .

(b)  $r(t) = \frac{\sin(7t)}{t^3}$ .

3. Consider the following 3 vectors:

$$\vec{a} = \langle 1, 2, 3 \rangle \quad \vec{b} = \langle 0, -1, 1 \rangle \quad \vec{c} = \langle 2, 4, 6 \rangle$$

(a) Compute  $\vec{a} + \vec{b}$ .

(b) Compute  $\vec{b} - 3\vec{c}$ .

(c) Compute  $\|\vec{a} - \vec{b}\|$ .

(d) Compute  $\vec{b} \cdot \vec{c}$ .

4. <sup>3</sup> Answer the following questions about parallel and perpendicular vectors.

(a) For what values of  $k$  (if any) are the vectors  $\vec{a} = \langle 1, k, 3 \rangle$  and  $\vec{b} = \langle 2, -3, 4 \rangle$  perpendicular?

(b) For what values of  $q$  (if any) are the vectors  $\vec{c} = \langle q, 2 \rangle$  and  $\vec{d} = \langle 2, 3 \rangle$  parallel?

5. Find all the unit vectors that are perpendicular to the vector  $\langle -3, 4 \rangle$ .

Follow-up: for the special case of 2D vectors, how can you quickly build a perpendicular vector to  $\langle a, b \rangle$ ?

6. Consider the vector  $\vec{v} = \langle 10, -4, 73 \rangle$ . *Without any calculation*, find a vector  $\vec{w}$  for which  $\vec{v} \cdot \vec{w} = 0$ .

---

<sup>1</sup>Sources: previous APSC 171 Test 1

<sup>2</sup>Source: previous APSC 171 Test 1

<sup>3</sup>Source: previous APSC 171 Test 1